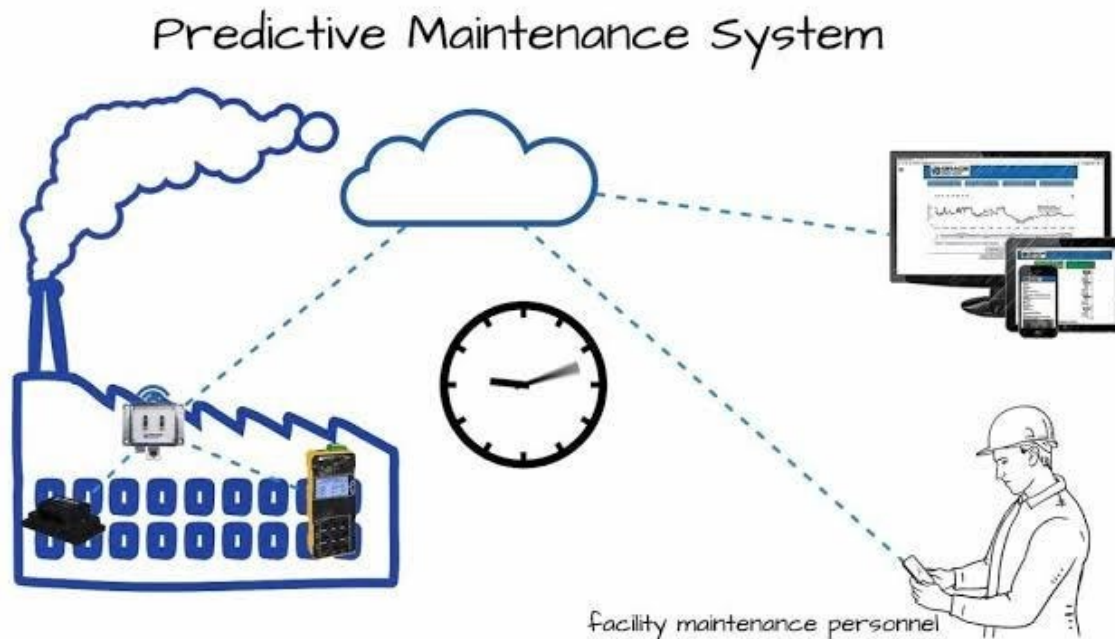


Predictive Maintenance using Supervised Machine Learning

Build a supervised Machine Learning model that predicts machine temperature or failure conditions using IoT sensor data.



- Use IoT sensor datasets
- Prepare AI-ready data
- Train an ML model
- Perform predictions
- Evaluate model accuracy
- Simulate AIoT intelligence pipeline

Concepts Covered

Supervised Learning

Features & Labels

Train/Test Split

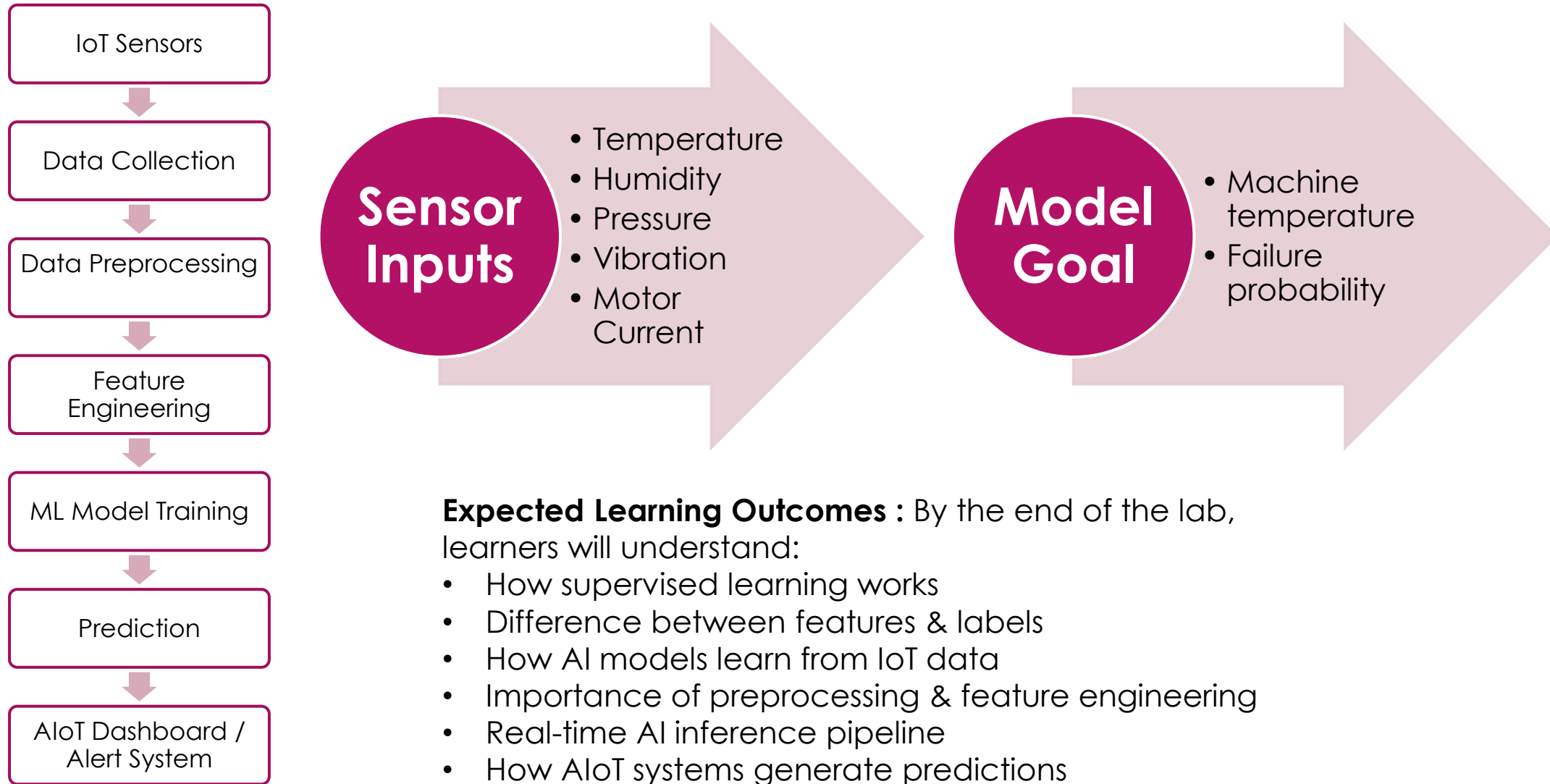
Regression vs Classification

Model Training Prediction

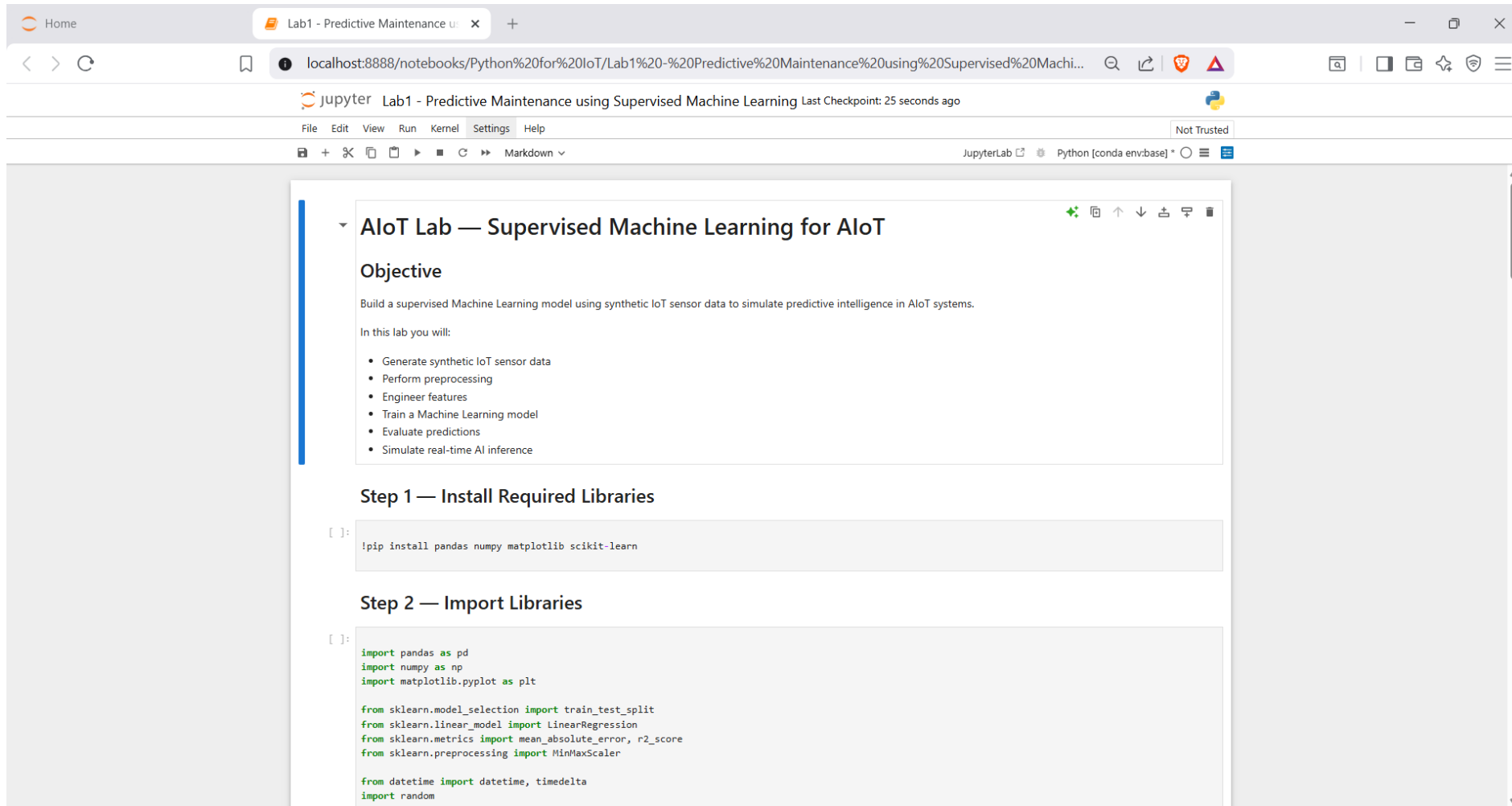
Evaluation Metrics AIoT

Inference Pipeline

Predictive Maintenance using Supervised Machine Learning



Predictive Maintenance using Supervised Machine Learning



The screenshot shows a JupyterLab interface in a web browser. The browser tab is labeled 'Lab1 - Predictive Maintenance using Supervised Machine Learning'. The address bar shows the URL 'localhost:8888/notebooks/Python%20for%20IoT/Lab1%20-%20Predictive%20Maintenance%20using%20Supervised%20Machi...'. The JupyterLab header shows the title 'Lab1 - Predictive Maintenance using Supervised Machine Learning' and 'Last Checkpoint: 25 seconds ago'. The left sidebar contains a file explorer with a folder named 'AloT Lab — Supervised Machine Learning for AloT'. The main area displays the notebook content, which includes an 'Objective' section, a list of tasks, and two code cells for installing and importing libraries.

AloT Lab — Supervised Machine Learning for AloT

Objective

Build a supervised Machine Learning model using synthetic IoT sensor data to simulate predictive intelligence in AloT systems.

In this lab you will:

- Generate synthetic IoT sensor data
- Perform preprocessing
- Engineer features
- Train a Machine Learning model
- Evaluate predictions
- Simulate real-time AI inference

Step 1 — Install Required Libraries

```
[ ]: !pip install pandas numpy matplotlib scikit-learn
```

Step 2 — Import Libraries

```
[ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
from sklearn.preprocessing import MinMaxScaler

from datetime import datetime, timedelta
import random
```